BAHIR DAR UNIVERSITY

BAHIR DAR INSTTUTE OF TECHNOLOGY

FACULTY OF COMPUTING

PROGRAM: BSc. IN COMPUTER SCIENCE

RESEARCH PROPOSAL

TITEL: THE EFFECT OF AI MODEL SIMULATION ON STUDENT ENGAGEMENT

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2025 GC

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Title: The Effect of AI Model Simulation on Student Engagement

1. Introduction and Background of the Study

Student engagement is a critical factor in determining the effectiveness of teaching and learning processes, academic achievement, and student retention in higher education. Engagement is commonly conceptualized as a multidimensional construct encompassing behavioral, emotional, and cognitive components (Fredricks, Blumenfeld, & Paris, 2004). High levels of student engagement are associated with improved learning outcomes, deeper understanding of course material, and greater academic persistence (Kuh, 2009).

In computer science education, particularly in abstract and conceptually complex topics, traditional lecture-based instruction often fails to sustain students' active participation and interest. As a result, learners may experience disengagement, reduced motivation, and superficial understanding of core concepts (Schindler et al., 2017).

Recent advances in Artificial Intelligence (AI) have introduced innovative educational tools such as AI model simulations, intelligent tutoring systems, and conversational AI assistants. These tools offer interactive, adaptive, and personalized learning experiences that align with student-centered and constructivist learning theories (Luckin et al., 2016; Woolf, 2010). AI model simulations, in particular, enable students to visualize abstract processes, experiment with models, and receive immediate feedback, thereby promoting active learning and engagement.

International organizations and researchers have highlighted the growing potential of AI technologies to transform educational practices and enhance learner engagement (OECD, 2021; Zawacki-Richter et al., 2019). However, empirical evidence on the effectiveness of AI model simulations in improving student engagement remains limited, especially in developing countries and within computer science education contexts.

Therefore, this study seeks to investigate the effect of AI model simulation on student engagement, with the aim of providing empirical evidence to inform instructional practices and educational technology adoption in higher education

2. Statement of the Problem and Justification

Despite the increasing adoption of AI-based tools in educational settings, their integration is often driven by technological enthusiasm rather than rigorous empirical evaluation. Many institutions implement AI model simulations without sufficient evidence regarding their comparative effectiveness against traditional teaching methods, particularly in terms of student engagement (Zawacki-Richter et al., 2019).

Student engagement is a foundational determinant of successful learning; yet, many computer science courses continue to rely heavily on traditional instructional approaches that may not adequately address diverse learning needs or promote sustained engagement (Schindler et al., 2017). Without empirical studies evaluating AI model simulations, educators and policymakers lack reliable guidance on whether such tools genuinely enhance engagement or simply introduce technological complexity.

The absence of evidence-based evaluation poses risks, including ineffective instructional design, misallocation of educational resources, and missed opportunities to improve learning experiences through personalized and interactive approaches (Pane et al., 2015; OECD, 2021).

This study is justified because it aims to:

- Provide empirical evidence on the effect of AI model simulation on student engagement (Fredricks et al., 2004).
- Support evidence-based decision-making regarding the integration of AI technologies in higher education (Kuh, 2009; OECD, 2021).
- Contribute to the growing body of knowledge in educational technology and AI-supported learning (Luckin et al., 2016; Zawacki-Richter et al., 2019).

3. Objectives of the Study

3.1 General Objective

To examine the effect of AI model simulation on student engagement in education.

3.2 Specific Objectives

1. To assess the level of student participation when AI model simulations are used in teaching (Fredricks et al., 2004).
2. To compare student engagement levels between AI-supported instruction and traditional teaching methods (Pane et al., 2015).
3. To identify which dimensions of student engagement (behavioral, emotional, cognitive) are most influenced by AI model simulation (Schindler et al., 2017).
4. To analyze students' perceptions of AI model simulation as a learning support tool (Woolf, 2010).

4. Hypothesis/Research Questions

4.1 Research Questions

1. How deeply involved are students in learning when a simulated AI model is incorporated into teaching?
2. Is there a noticeable gap in the level of learning engagement for students who are taught utilizing AI model simulation versus those who are taught through conventional means?
3. What dimensions of student engagement are impacted the most by AI model simulation?
4. What are students' opinions regarding the helpfulness and ease of use of AI model simulation in the learning process?

4.2 Research Hypotheses

H1: Students in the experimental group (exposed to AI model simulation) will demonstrate a significantly higher level of overall engagement than students in the control group (traditional instruction).

H2: AI model simulation will have a significant positive effect on the behavioral, emotional, and cognitive engagement of students.

H0: There is no significant difference in student engagement between students taught using AI model simulation and those taught using traditional methods.

5. Scope or Delimitation of the Study

- Population & Sample: Selected universities' undergraduate students via introductory courses in programming and computer science.
- Content Scope: Module-based integration of specific course parts with AI model simulators (Conversation AI problem-solving aide, code-tracing simulators)

Delimitations:

1. Long-term academic performance will not be considered, as the focus of this study will be on engagement as the dependent variable.
2. The context of the findings will be specific to computer science education, and the AI tools used in this study.
3. Some extraneous variables may be beyond control, owing to the nature of the quasi-experimental design.

6. Methodology of the Study

6.1. Research Design

This study will utilize a quasi-experimental approach, adopting a causal (explanatory) design. The study will use two groups:

- Experimental group: students who will be taught with an AI model simulation.
- Control group: students who will be taught with traditional methods of instruction.

6.2. Research Approach

A mixed-methods approach will be used, as follows:

- Measuring student engagement will utilize quantitative methods.
- For capturing and understanding the students' perceptions and experiences, qualitative methods will be used.

6.3 Population and Sample

- Target Population: Computer Science undergraduate students attending programming courses at Bahir Dar University.
- Sample: A purposive sampling technique will be used involving intact classes.

8. Budget Schedule

Item	Estimated Cost (USD)	Justification

A. Software and subscription

AI simulation platform license	400	Educational discount for one semester use
Data analysis software (SPSS/NVivo)	300	Annual academic license
Cloud storage & backup	100	Secure data storage and backup

B. Participant Compensation

Student incentive vouchers	600	$10 vouchers for 60 participants to encourage completion

C. Research Materials

Printing & stationery	150	Questionnaires, consent forms, observation sheets

D. Personnel

Research assistant stipend	800	Part-time assistance for data collection and entry
Interview transcription services	200	Professional transcription of qualitative data

E. Dissemination (Optional)

Conference registration fee	350	For presenting findings at academic conference
Open-access publication fee	500	If targeting open-access journal

F. Miscellaneous & Contingency

Unforeseen expenses	300	Technical issues, additional materials, or timeline extension
TOTAL ESTIMATED BUDGET	3700	(Core required: $2,550; Optional dissemination: $850)

9. Time Schedule

Phase	Activity	Duration (Months)
I. Preparatory Phase	Proposal approval & ethical clearance	1

	Literature review & instrument design	2
	Pilot testing & refinement	1
II. Implementation Phase	Participant recruitment & pre-test	1
	AI simulation intervention delivery	3
	Post-test data collection & interviews	1
III. Analysis & Reporting	Data analysis (quantitative & qualitative)	1
	Report writing & finalization	2
	TOTAL DURATION	12

10. Reference

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